

REFERENCE



The **id** Connection

One name links HTML · CSS · JS

id 하나로 세 가지를 연결

BIG IDEA

One **id**, three layers

Give an element an **id**, and all three layers can find it.



HTML

name it

```
id="hero"
```



CSS

style it

```
#hero { }
```



JS

grab it

```
getElementById("hero")
```

같은 id로 HTML·CSS·JS 연결

Name the element

```
<h1 id="hero">Hello</h1>
```

The **id** is a name tag. It must be **unique** — only one element per page can use it.

id = 이름표 (한 페이지에 하나만)

2 · CSS

Style it with

#



```
#hero {  
  color: tomato;  
  font-size: 48px;  
}
```

means "find the id". #hero targets id="hero".

#는 id를 가리킴

3 · JS

Grab it with `getElementById` ⚡

```
let el = document.getElementById("hero");  
el.textContent = "Changed by JavaScript!";
```

No `#` here — just the name `"hero"`. JS finds the element, then you change it.

JS는 #없이 이름만 / 요소를 잡아서 바꿈

ALL TOGETHER

Same **id**, three places

```
<h1 id="hero">Hello</h1>  <!-- name it -->

<style>
  #hero { color: tomato; }  /* style it */
</style>

<script>
  let el = document.getElementById("hero");  // grab it
  el.textContent = "Hi!";
</script>
```

The same word **hero** connects all three.

⚠ WATCH

The id must match exactly



Works

```
id="hero"  
getElementById("hero")
```



Broken

```
id="hero"  
getElementById("Hero")
```

Same spelling, same **capital letters**. Hero ≠ hero.

철자·대소문자까지 똑같이

DEEPER

JS changes the **style**, not just text

```
// 1. change one style directly
```

```
el.style.color = "tomato";
```

```
// 2. or flip a whole CSS class on
```

```
el.classList.add("active");  /* CSS: .active { ... } */
```

CSS decides the **look**. JS decides the **when**.

CSS는 모양, JS는 언제 바뀔지

PATTERN

Make it wiggle ✨

```
/* CSS – describe the move */  
@keyframes wiggle { 50% { transform: rotate(8deg); } }  
.wiggle { animation: wiggle .3s; }  
  
// JS – switch it on with a click  
btn.onclick = () => box.classList.add("wiggle");
```

This is the **pattern** for every challenge: CSS the look, JS adds the class.

CSS로 동작 정의 → JS로 클래스 추가

YOUR TURN!

Try it yourself

✓ Add `id="title"` to a heading

✓ Style `#title` in CSS (pick a color)

✓ Grab it with `getElementById("title")`

✓ Change its `textContent`

세 군데 모두 같은 id 사용

PRACTICE



Build these step by step

Follow along first. Then fly solo.

Every one is the **same 3 steps**: **1)** HTML — give it an `id` · **2)** CSS — write a class · **3)** JS — add the class on click.

전부 같은 3단계 흐름

Change colour on click

Step 1 — HTML (name it + a button)

```
<p id="box">Tap me</p>
<button id="btn">Paint</button>
```

Step 2 — CSS (a colour class)

```
.red { color: tomato; }
```

Step 3 — JS (on click, add the class)

```
let box = document.getElementById("box");
document.getElementById("btn").onclick = () =>
  box.classList.add("red");
```

Make it grow

Step 1 — HTML (name it)

```
<div id="box">Grow me</div>
```

Step 2 — CSS (a class that scales up)

```
.big { transform: scale(1.3); }
```

Step 3 — JS (add it when clicked)

```
let box = document.getElementById("box");  
box.onclick = () => box.classList.add("big");
```

흐름은 똑같고 클래스만 바뀜

Count the clicks

Step 1 — HTML (a button showing 0)

```
<button id="box">0</button>
```

Step 2 — JS (keep a count, +1 each click)

```
let box = document.getElementById("box");  
let count = 0;  
box.onclick = () => {  
  count = count + 1;  
  box.textContent = count;  
};
```

No CSS here — JS changes the `textContent`.

NOW YOU TRY

Your turn — make it...

Same 3 steps. Pick one, build all three layers.



glow on hover

CSS only · `#box:hover`



fade out

`opacity: 0` + `transition`



spin

`transform: rotate(360deg)`



drop a shadow

`box-shadow` `class`



change the text

`textContent`



make it bigger

`font-size` `class`

같은 3단계: id → CSS → JS

BOSS CHALLENGE

Level up



Toggle — click on, click again off (`classList.toggle`)



Two at once — wiggle **and** recolour on one click



Counter + colour — every 5 clicks, change the colour



Invent your own — pick a "make it ___" and build all 3 steps

한 번의 클릭으로 여러 가지



One id, all connected!

HTML names it · CSS styles it · JS changes it.

잘했어요!